

AIM:

To visualize Baseball Databank data using Python and create trend, category, distribution and relationship visualizations.

PROCEDURE:

1. Load the Baseball batting and fielding datasets.
2. Calculate total runs for the recent 24 seasons.
3. Create a line chart for recent season runs.
4. Calculate total runs by league and visualize them using a bar chart.
5. Visualize the most common fielding positions using a pie chart.
6. Create a histogram to show the frequency distribution of Runs.
7. Create a scatter plot to study batting relationships.

OUTPUT:

```
In [1]: from pathlib import Path

def find_data(filename):
    candidates = [
        Path(filename),
        Path("data") / filename,
        Path("../data") / filename,
        Path("../..data") / filename,
        Path("/content") / filename,
    ]
    for p in candidates:
        if p.exists():
            return p
    raise FileNotFoundError(f"{filename} not found. Upload the Baseball Databank CSV or keep it in the repository data folder.")

import pandas as pd
import matplotlib.pyplot as plt

batting = pd.read_csv(find_data("Batting.csv"))
fielding = pd.read_csv(find_data("Fielding.csv"))

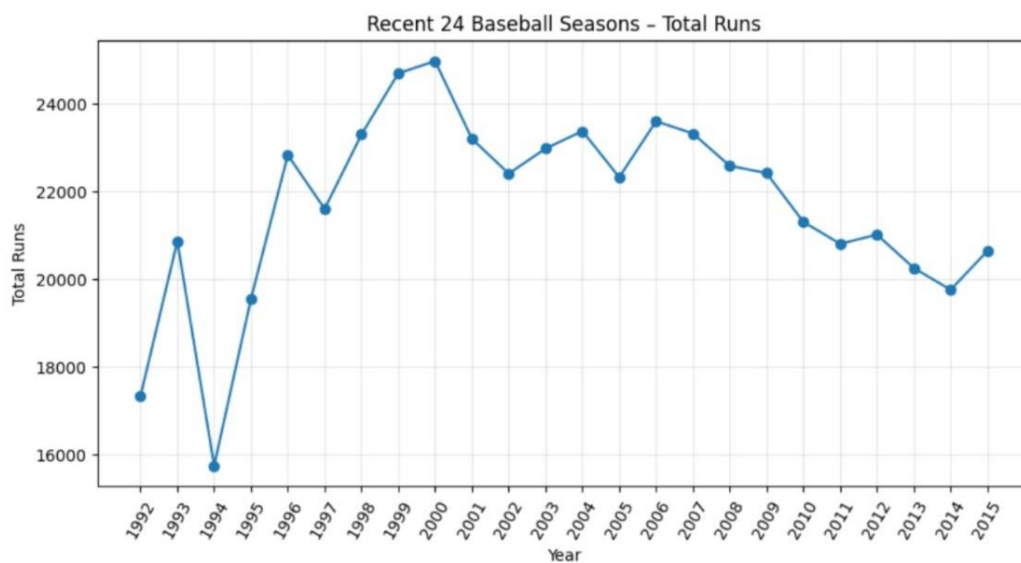
print("Baseball datasets loaded successfully")
print("Rows:", len(batting))
print("Columns:", len(batting.columns))

Baseball datasets loaded successfully
Rows: 101332
Columns: 22
```

```
In [2]: yearly = batting.groupby("yearID")["R"].sum().tail(24)
yearly
```

```
Out[2]: yearID
1992    17341.0
1993    20864.0
1994    15752.0
1995    19554.0
1996    22831.0
1997    21604.0
1998    23297.0
1999    24691.0
2000    24971.0
2001    23199.0
2002    22408.0
2003    22978.0
2004    23376.0
2005    22325.0
2006    23599.0
2007    23322.0
2008    22585.0
2009    22419.0
2010    21308.0
2011    20808.0
2012    21017.0
2013    20255.0
2014    19761.0
2015    20647.0
Name: R, dtype: float64
```

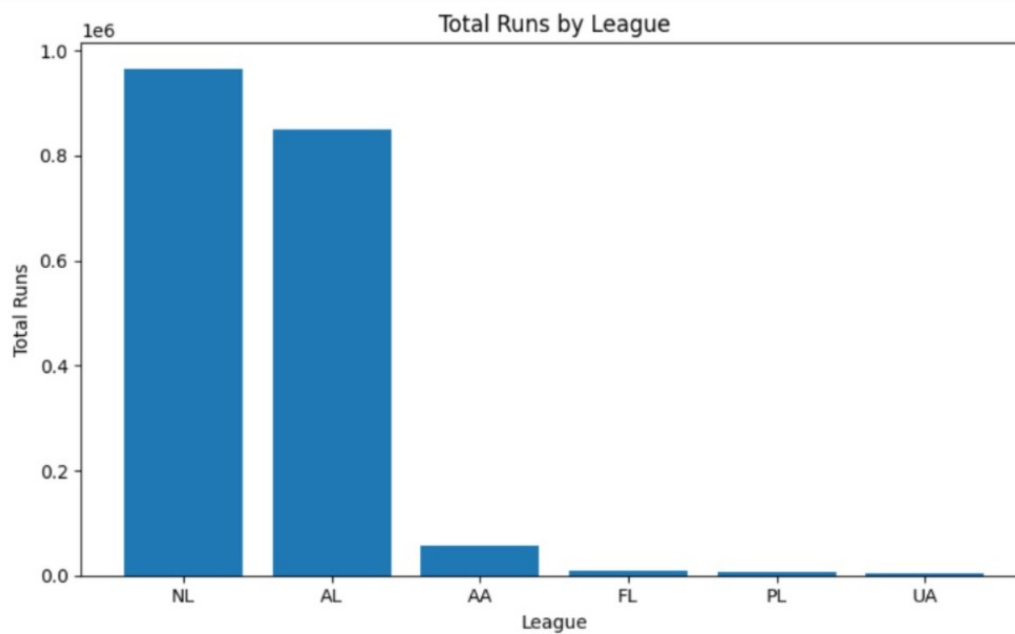
```
In [3]: plt.figure(figsize=(9,5))
plt.plot(yearly.index.astype(str), yearly.values, marker="o")
plt.title("Recent 24 Baseball Seasons - Total Runs")
plt.xlabel("Year")
plt.ylabel("Total Runs")
plt.xticks(rotation=60)
plt.grid(True, alpha=.3)
plt.tight_layout()
plt.show()
```



```
In [4]: league_runs = batting.groupby("lgID")["R"].sum().sort_values(ascending=False)
league_runs
```

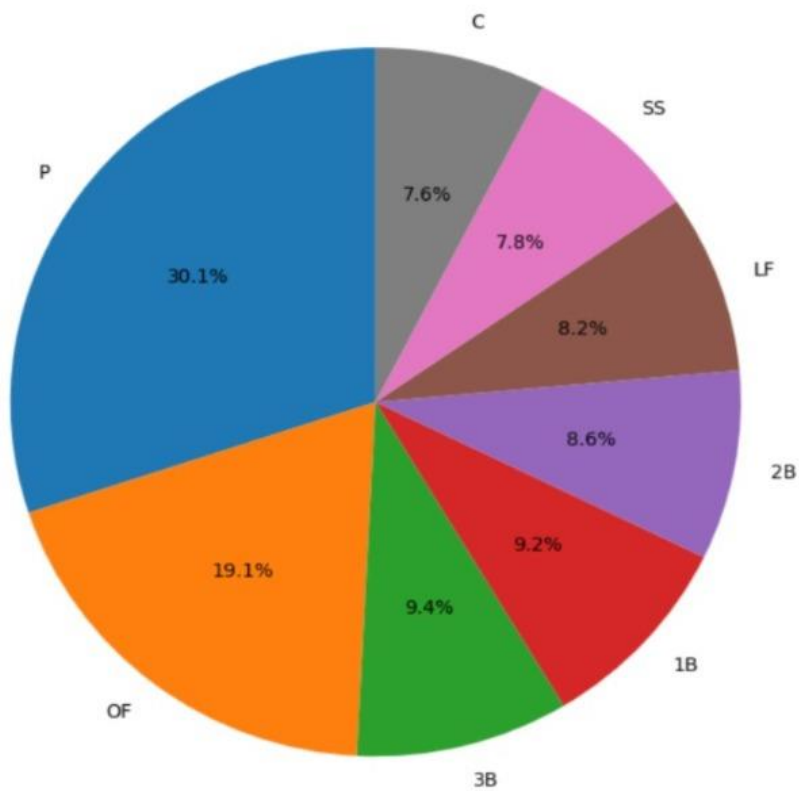
```
Out[4]: lgID
NL      966403.0
AL      849755.0
AA       57307.0
FL        9894.0
PL        7278.0
UA        4825.0
Name: R, dtype: float64
```

```
In [5]: plt.figure(figsize=(8,5))
plt.bar(league_runs.index.astype(str), league_runs.values)
plt.title("Total Runs by League")
plt.xlabel("League")
plt.ylabel("Total Runs")
plt.tight_layout()
plt.show()
```

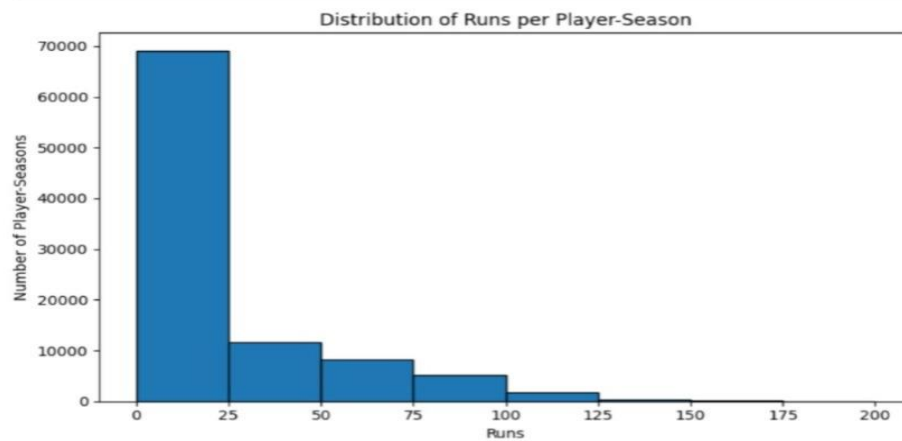


```
In [6]: positions = fielding["POS"].value_counts().head(8)
plt.figure(figsize=(7,7))
plt.pie(positions.values, labels=positions.index, autopct="%1.1f%%", startangle=90)
plt.title("Fielding Position Distribution")
plt.tight_layout()
plt.show()
```

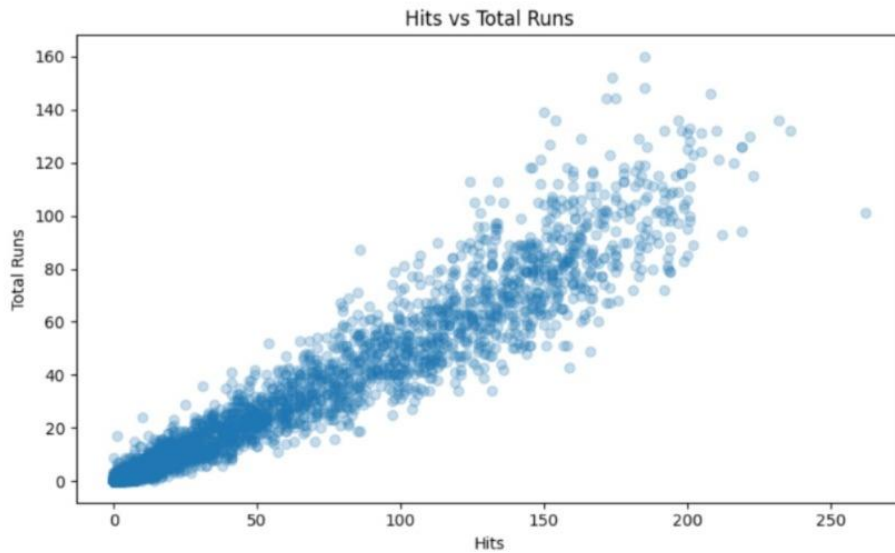
Fielding Position Distribution



```
In [7]: plt.figure(figsize=(8,5))
plt.hist(batting["R"].dropna(), bins=range(0, int(batting["R"].max()) + 25, 25), edgecolor="black")
plt.title("Distribution of Runs per Player-Season")
plt.xlabel("Runs")
plt.ylabel("Number of Player-Seasons")
plt.tight_layout()
plt.show()
```



```
In [8]: sample = batting.sample(min(5000, len(batting)), random_state=42)
plt.figure(figsize=(8,5))
plt.scatter(sample["H"], sample["R"], alpha=.25)
plt.title("Hits vs Total Runs")
plt.xlabel("Hits")
plt.ylabel("Total Runs")
plt.tight_layout()
plt.show()
```



Result:

Thus, the Baseball Databank was successfully visualized using trend, league, fielding-position, histogram and scatter plots.